Question	Answer	Marks	Guidance
2(a)	resultant force (in any direction) is zero	B1	Allow 'net force is zero' Allow 'sum of forces is zero' or 'total force is zero' or '$\Sigma F = 0$' Allow 'sum of vertical forces is zero <u>and</u> sum of horizontal forces is zero'. Allow 'resultant force about any point is zero' Ignore 'forces balance' or 'forces cancel' or 'forces are equal' Ignore just 'no (external) forces act' or just 'force is zero' Ignore object stationary / constant velocity / zero acceleration etc
	resultant torque / moment (about any point) is zero	B1	Allow 'net moment is zero' Allow 'sum of moments is zero' or 'total moment is zero' Allow 'sum of CW moments = sum of ACW moments' Allow 'sum of CW and ACW moments is zero' Allow 'resultant moment at a point is zero' Allow 'torque' instead of 'moment' throughout Ignore 'moments balance' or 'moments cancel' Ignore 'no moments' or 'moment is zero' Not 'momentum' instead of 'moment'
2(b)(i)	one equation for forces on **either** block **or** plank: $2T\cos\theta = R + W_P$ or $2T\cos\theta + R = W_B$	M1	A full equation for the forces on either object must be seen Allow $0 = R + W_P - 2T\cos\theta$ Allow $0 = W_B - R - 2T\cos\theta$ Ignore just $R + W_P - 2T\cos\theta = W_B - R - 2T\cos\theta$ Allow $\sin(90 - \theta)$ for $\cos\theta$ Not $\sin\theta$ for $\cos\theta$
	both equations for forces on block **and** plank **and** $(2T\cos\theta =)\ R + W_P = W_B - R$	A1	Second mark for both separate equations seen and 'show that' answer given. <u>Special case:</u> $2T\sin\theta = R + W_P$ <u>and</u> $2T\sin\theta + R = W_B$ <u>and</u> $(2T\sin\theta =)\ R + W_P = W_B - R$ credit 1 / 2

Question	Answer	Marks	Guidance
2(b)(ii)	$2R = 8.1 \times 9.81 - 4.5 \times 9.81$	C1	Allow $2R = 8.1g - 4.5g$ Allow $g = 10$
	$R = 18$ N	A1	
2(c)(i)	Hooke's (law)	A1	Allow phonetic misspellings of 'Hooke' e.g. Hook, Huk, Hookie
2(c)(ii)	When the tension is removed, the ropes have original length / extension is zero OR when the tension is removed object is not deformed	A1	Allow 'it' to mean ropes Allow 'force' / 'load' / 'T' / 'stress' / weight' for tension Ignore 'mass' for tension Allow 'unloaded' for tension is removed Allow 'tension is released' (BOD) but ignore 'object released' Allow 'unstretched' / 'initial' for original Allow 'shape <u>and</u> size' for length